



2003
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown on every question

Total marks – 84

- Attempt Questions 1-7
- All questions are of equal value
- Start each question in new writing booklet

Question 1 (12 marks)

(a) Evaluate $\lim_{x \rightarrow 0} \frac{\tan 4x}{x}$. **1**

(b) Find $\frac{d}{dx}(2x^3e^{3x})$. **2**

(c) Solve $\frac{1}{2-x} > 3$. **3**

(d) State the domain and range of the function **2**

$$f(x) = 2 \cos^{-1}\left(\frac{x}{3}\right)$$

(e) Find the acute angle between the lines **2**

$$\begin{aligned} y &= 3x - 5 \\ 2x + y - 7 &= 0 \end{aligned}$$

to the nearest degree.

(f) Evaluate $\int \frac{\cos x}{1 + 2 \sin x} dx$ **2**

Question 3 (12 marks)

- (a) Six people are seated in a straight line.
- (i) How many seating arrangements are possible? **1**
- (ii) How many arrangements are possible if Tarzan and Jane occupy the seats at either end? **2**
- (b) (i) Show that $x^3 + 2x - 17 = 0$ has a root between $x=2$ and $x=3$ **1**
- (ii) Using an approximation of $x = 2.4$, use one application of Newton's method to find a better approximation for this root. Give your answer to two decimal places. **3**
- (c) Use a table of standard integral to evaluate **2**

$$\int \frac{1}{\sqrt{x^2 + 9}} dx$$

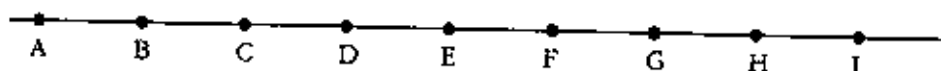
(d) $\int_0^{\frac{3}{4}} \frac{1}{9+16x^2} dx$

3

Question 4 (12 marks)

- (a) (i) In what ratio does I divide AG?

1



- (ii) W(2,3) divides XY internally in the ratio $k:l$ where $X(-1,1)$ and $Y(7,9)$. Find the ratio $k:l$. 2

- (b) The polynomial $P(x) = x^3 - 3x^2 + kx - 2$ has roots α, β, γ .

- (i) Find the value of $\alpha + \beta + \gamma$. 1

- (ii) Find the value of $\alpha\beta\gamma$. 1

- (iii) It is known that two roots are the reciprocal of each other. Find the value of the third root and hence find the value of k . 2

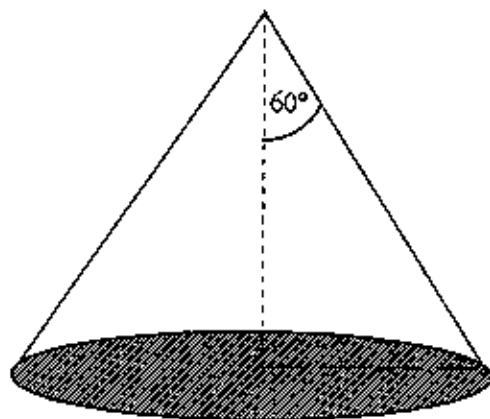
- (c) Marvin the Martian has a body temperature of 100°C . When Marvin sleeps his body temperature obeys Newton's Law of Cooling according to the law $\frac{dT}{dt} = k(T - A)$, where T is Marvin's body temperature and A is the temperature

of the surrounding air.

- (i) Show that $T = A + Ce^{kt}$, where C and k are constants, satisfies Newton's Law of Cooling. 2
- (ii) Marvin goes to sleep at 10 pm. His temperature at midnight is 95°C . Marvin's bedroom is air conditioned with the temperature set at 20°C . Assuming Marvin continues to sleep what will be his body temperature at 8am? 3

Question 5 (12 marks)

- (a) Use the principle of Mathematical Induction to show that $7^n + 13^n$ is divisible by 10 for n odd integers. 3
- (b) Sand pours onto the ground and forms a cone where the semi-vertical angle is 60° . The height of the cone at time t seconds is h cm and the radius of the base is r cm. Sand is being poured onto the pile at a rate of $12\text{cm}^3/\text{s}$.



- (i) Show that $r = \sqrt{3}h$ 1
- (ii) Find the rate at which the height is increasing at the instant when the height is 12 cm. 3

$$[\text{Volume of a cone} = \frac{1}{3}\pi r^2 h]$$

(c) Consider the function

$$f(x) = \tan^{-1} x + \tan^{-1} \left(\frac{1}{x} \right)$$

- | | | |
|-------|--|---|
| (i) | State any values of x for which $f(x)$ is undefined. | 1 |
| (ii) | Show that $f(1) = \frac{\pi}{2}$ | 1 |
| (iii) | Show that $f'(x) = 0$ | 2 |
| (iv) | Sketch the graph of $y = f(x)$ | 1 |

Question 6 (12 marks)

A particle moves in Simple Harmonic Motion with amplitude a , in the form $\ddot{x} = -4x$ where x is the displacement, in metres, from the origin O and t is the time in seconds.

- | | | |
|-------|---|---|
| (i) | Prove that $v^2 = 4(a^2 - x^2)$ | 3 |
| (ii) | The particle moves so that $x = 2$, $v = 4$ find the value of a . | 1 |
| (iii) | Find an expression for v in terms of displacement. | 2 |
| (iv) | By setting $v = \frac{dx}{dt}$ and taking the reciprocal, prove that $x = 2\sqrt{2} \sin 2t$
if when $t = \frac{\pi}{4}$, $x = 2\sqrt{2}$. | 3 |
| (v) | Where would you expect the maximum speed to occur? | 1 |
| (vi) | Hence, or otherwise, find the maximum speed of the particle. | 2 |

Question 7 (12 marks)

- (a) A particle moves according to the equation $x = 2e^{-t}(\cos t + \sin t)$.

It moves in the interval $0 \leq t \leq 2\pi$.

- | | | |
|-------|--|---|
| (i) | Show that $\dot{x} = -4e^{-t} \sin t$ and find the acceleration function $\ddot{x}$. | 2 |
| (ii) | Discuss the displacement as $t \rightarrow \infty$. | 1 |
| (iii) | Find the times when the particle is at the origin. | 2 |
| (iv) | When is the particle moving in the positive direction. | 1 |
| (v) | Find the times when the particle will be stationary. | 2 |
| (vi) | Find the displacement at the times when the particle is stationary, (give your answers correct to three decimal places). | 1 |
| (vii) | Draw a neat, full-page sketch of $x = 2e^{-t} (\cos t + \sin t)$, giving endpoints, stationary points and intercepts | 3 |

Kin Coppal - Rosebay (2003)

Trial Extension 1 solutions 2003

Question 1:

(a) $4 \lim_{x \rightarrow 0} \frac{\tan 4x}{4x} = 4$ ✓

(b) $2x^3 3e^{3x} + 6x^2 e^{3x} = 6x^2 e^{3x} (x+1)$ ✓ ✓

(c) $(2-x)^2 \times \frac{1}{2-x} \geq 3(2-x)^2 \quad x \neq 2$ ✓

$$2-x \geq 3(2-x)^2$$

$$2-x-3(2-x)^2 \geq 0$$

$$(2-x)(1-3(2-x)) \geq 0$$

$$(2-x)(3x-5) \geq 0$$

$2 < x \leq \frac{5}{3}$ never ever write like this ✓

(d) $-1 \leq \frac{x}{3} \leq 1$ ✓ $0 \leq f(x) \leq 2\pi$
 $-3 \leq x \leq 3$ ✓

(e) $m_1 = 3$ ✓
 $m_2 = -2$

For an acute angle

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan \theta = \left| \frac{3 - (-2)}{1 - 6} \right|$$

$$\tan \theta = \left| \frac{5}{-5} \right| = 1$$

$$\theta = 45^\circ$$
 ✓

(f) $\frac{1}{2} \log(1+2\sin x) + C$ ✓ ✓

Question 2

(a) $\frac{du}{dx} = 4x \quad x=2 \quad u=9$ ✓
 $x=0 \quad u=1$

$$dx = \frac{du}{4x}$$

$$\int_0^2 \frac{8x}{\sqrt{1+2x^2}} dx = \int_1^9 \frac{8x}{\sqrt{u}} \frac{du}{4x} = \int_1^9 2u^{-\frac{1}{2}} du = \left[4u^{\frac{1}{2}} \right]_1^9 = 12 - 4 = 8$$
 ✓

(b) $\tan x = \frac{1}{\sqrt{3}}$ ✓

$$x = n\pi + \tan^{-1} \frac{1}{\sqrt{3}}$$

$$x = n\pi + \frac{\pi}{6}$$
 ✓

(c)

$$P(2) = 2(2)^4 - 4(2)^3 + 4(2)^2 - 15(2) + 14 = 0$$

∴ $(x-2)$ is a factor via the factor theorem

(d)

$$\cos 2x = 2\sin^2 x - 1$$

$$\text{so } \cos 4x = 2\sin^2 2x - 1$$

$$\sin^2 2x = \frac{1}{2}(\cos 4x + 1)$$
 ✓

$$\int_0^{\frac{\pi}{4}} \sin^2 2x dx = \frac{1}{2} \int_0^{\frac{\pi}{4}} (\cos 4x + 1) dx$$

$$= \frac{1}{2} \left[\frac{\sin 4x}{4} + x \right]_0^{\frac{\pi}{4}}$$
 ✓

$$= \frac{1}{2} \left[\left(\frac{\sin \pi}{4} + \frac{\pi}{4} \right) - 0 \right]$$

$$= \frac{\pi}{8}$$
 ✓

- (e) (i) Two tangents from an exterior point to a circle are equal. ✓

(ii)

Since $\angle BFE = \alpha$, then $\angle FBE = \alpha$ (isos Δ) ✓

Since $\angle BDE = \beta$, then $\angle DBE = \beta$ (isos Δ) ✓

In ΔBFD $\alpha + \alpha + \beta + \beta = 180$ ($\angle$ sum of Δ)

$$2\alpha + 2\beta = 180$$

$$\alpha + \beta = 90 \quad \checkmark$$

$$\therefore \angle FBD = 90^\circ$$

Question 3.

(a) (i) $6! = 720$ ✓

(ii) $4 \times 2! = 48$ ✓

(b) (i)

$$f(2) = (2)^3 + 2(2) - 17$$

$$= 8 + 4 - 17$$

$$= -5$$

$$f(3) = (3)^3 + 2(3) - 17$$

$$= 16$$

Since $f(x)$ changes sign between $x=2$ and $x=3$ and since $f(x)$ is continuous for all x , $f(x)$ must be zero somewhere between $x=2$ and $x=3$.

(ii)

$$f'(x) = 3x^2 + 2$$

$$f'(2.4) = 3(2.4)^2 + 2 = 19.24 \quad \checkmark$$

$$f(2.4) = (2.4)^3 + 2(2.4) - 17 = 1.624$$

$$x_2 = x_1 - \frac{f(x)}{f'(x)}$$

$$= 2.4 - \frac{1.624}{19.24} \quad \checkmark$$

$$= 2.3155925 \dots$$

$$= 2.32 \text{ (2 d.p.)} \quad \checkmark$$

(c) $\ln(x + \sqrt{x^2 + 9}) + C$ ✓

$$\left[\frac{1}{16} \times \frac{4}{3} \tan^{-1} \frac{4x}{3} \right]_0^3 \quad \checkmark$$

(d) $= \frac{1}{16} \times \frac{4}{3} \tan^{-1} 1 - \frac{1}{16} \times \frac{4}{3} \tan^{-1} 0$

$$= \frac{1}{16} \times \frac{4}{3} \times \frac{\pi}{4}$$

$$= \frac{\pi}{48} \quad \checkmark$$

Question 4.

(a) (i) $-4:1$ ✓

(ii) $x = \frac{kx_2 + lx_1}{k+l}$

$$2 = \frac{7k-l}{k+l} \quad \checkmark$$

$$2k+2l = 7k-l$$

$$-5k = -3l$$

$$\frac{k}{l} = \frac{3}{5}$$

$$k:l = 3:5 \quad \checkmark$$

bad question.
use $y = \frac{kx+l}{k+l}$
and you get
different answer

This suggests that...
 $\times$ WY is not a line

(b) (i) $\alpha + \beta + \gamma = 3$ ✓

(ii) $\alpha\beta\gamma = 2$ ✓

(iii) $\alpha \times \frac{1}{\alpha} \times \gamma = 2$

$$\gamma = 2 \quad \checkmark$$

$$\alpha + \beta + 2 = 3$$

$$\alpha + \beta = 1$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = -k$$

$$1 + 2\beta + 2\alpha = -k$$

$$1 + 2(\beta + \alpha) = -k$$

$$1 + 2 \times 1 = -k$$

$$k = -3 \quad \checkmark$$

(c) (i)

$$T = A + Ce^{kt}$$

$$\frac{dT}{dt} = kCe^{kt} \quad \checkmark$$

$$\frac{dT}{dt} = k(T - A) \text{ as } Ce^{kt} = T - A \quad \checkmark$$

(ii)

$$\text{When } t = 0 \quad T = 100$$

$$100 = 20 + Ae^0$$

$$A = 80 \quad \checkmark$$

$$T = 20 + 80e^{kt}$$

$$\text{when } t = 2 \quad T = 95$$

$$95 = 20 + 80e^{k \cdot 2}$$

$$e^{2k} = \frac{15}{16}$$

$$2k = \ln \frac{15}{16}$$

$$k = \frac{1}{2} \ln \frac{15}{16} \quad \checkmark$$

$$\text{when } t = 10$$

$$T = 20 + 80e^{\frac{1}{2} \ln \frac{15}{16} \cdot 10}$$

$$T = 77.93571472$$

$$T = 78^\circ \quad \checkmark$$

Question 5:

(a) Let $7^n + 13^n = 10M$ where M is any integer.

For $n=1$ $7^1 + 13^1 = 20 = 10 \times 2$ $\checkmark$
which is divisible by 10.

Assume $7^n + 13^n = 10M$ is true for $n=k$
Prove true for $n=k+2$

$$13^k = 10M - 7^k$$

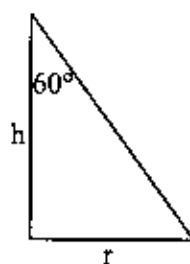
$$7^{k+2} + 13^{k+2} =$$

$$\begin{aligned} 7^2 7^k + 13^2 13^k &= 7^2 7^k + 13^2 (10M - 7^k) \quad \checkmark \\ &= 49 \cdot 7^k + 1690M - 169 \cdot 7^k \\ &= 1690M - 120 \cdot 7^k \\ &= 10(169 - 12 \cdot 7^k) \quad \checkmark \end{aligned}$$

which is a multiple of 10, therefore true for $n=k+2$.

Since it is true for $n=1$, it is true for $n=1+2$
And so on, so it is true for all positive odd integers.

(b)



$$(i) \quad \tan 60^\circ = \frac{r}{h}$$

$$h \tan 60^\circ = r \quad \checkmark$$

$$r = \sqrt{3}h$$

$$(ii) \quad \frac{dV}{dt} = 12$$

$$V = \frac{1}{3} \pi r^2 h$$

$$V = \frac{1}{3} \pi (3h^2) h$$

$$V = \pi h^3 \quad \checkmark$$

$$\frac{dV}{dh} = 3\pi h^2$$

$$\frac{dV}{dh} = 432\pi \text{ when } h=12 \quad \checkmark$$

$$\frac{dh}{dt} = \frac{dh}{dV} \cdot \frac{dV}{dt}$$

$$= \frac{1}{432\pi} \cdot 12$$

$$= \frac{1}{36\pi} \text{ cm/s} \quad \checkmark$$

(c) (i) $x \neq 0$ ✓

$$f(1) = \tan^{-1} 1 + \tan^{-1} \left(\frac{1}{1} \right)$$

(ii) $= \frac{\pi}{4} + \frac{\pi}{4}$ ✓

$$= \frac{\pi}{2}$$

(iii)

$$f'(x) = \frac{1}{1+x^2} + \frac{1}{1+\left(\frac{1}{x}\right)^2} \cdot \frac{d}{dx} (x^{-1})$$

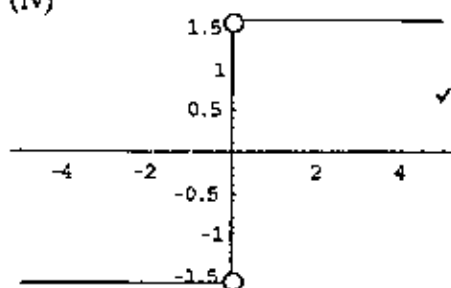
$$= \frac{1}{1+x^2} + \frac{1}{\frac{x^2+1}{x^2}} \cdot -x^{-2}$$
 ✓

$$= \frac{1}{1+x^2} + \frac{x^2}{x^2+1} \cdot -\frac{1}{x^2}$$

$$= \frac{1}{1+x^2} - \frac{1}{x^2+1}$$

$$= 0$$
 ✓

(iv)



Question 6 (12 marks) (Start a new booklet)

(i) $\ddot{x} = -4x$

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = -4x \quad \checkmark$$

$$\frac{1}{2}v^2 = -\frac{4x^2}{2} + C \quad (v=0, x=a)$$

$$0 = -2a^2 + C$$

$$C = 2a^2 \quad \checkmark$$

$$\frac{1}{2}v^2 = -2x^2 + 2a^2$$

$$v^2 = 4(a^2 - x^2) \quad \checkmark$$

(ii) $v^2 = 4(a^2 - x^2) \quad x=2, v=4$

$$16 = 4(a^2 - 4)$$

$$a^2 - 4 = 4$$

$$a^2 = 8$$

$$a = 2\sqrt{2} \quad (a > 0) \quad \checkmark$$

(iv) $v = 2\sqrt{8-x^2}$

$$\frac{dx}{dt} = 2\sqrt{8-x^2}$$

$$\frac{dt}{dx} = \frac{1}{2\sqrt{8-x^2}} \quad \checkmark$$

$$t = \frac{1}{2} \sin^{-1}\left(\frac{x}{2\sqrt{2}}\right) + C, \left(t = \frac{\pi}{4}, x = 2\sqrt{2}\right) \quad \checkmark$$

$$\frac{\pi}{4} = \frac{1}{2} \sin^{-1}\left(\frac{2\sqrt{2}}{2\sqrt{2}}\right) + C$$

$$\frac{\pi}{4} = \frac{1}{2} \sin^{-1}(1) + C$$

$$\frac{\pi}{4} = \frac{\pi}{4} + C$$

$$C = 0$$

$$t = \frac{1}{2} \sin^{-1}\left(\frac{x}{2\sqrt{2}}\right) \quad \checkmark$$

$$\sin(2t) = \frac{x}{2\sqrt{2}}$$

$$x = 2\sqrt{2} \sin(2t)$$

(v) at the origin (the centre of the motion). $\checkmark$

(iii) $v^2 = 4(8-x^2)$

$$v = \pm 2\sqrt{8-x^2} \quad \checkmark$$

but if $x=2, v=4 > 0$

$$v = 2\sqrt{8-x^2} \quad \checkmark$$

↳ nope.

SHM so it's $\pm$

$$(vi) \quad v = 4\sqrt{2} \cos(2t) \quad \boxed{\checkmark}$$

$$\begin{aligned} \text{max speed} &= 4\sqrt{2} \times 1 \\ &= 4\sqrt{2} \text{ m/s} \end{aligned} \quad \boxed{\checkmark}$$

Question 7 (12 marks) (Start a new booklet)

$$(i) \quad x = 2e^{-t}(\cos t + \sin t)$$

$$\dot{x} = (\cos t + \sin t) \times -2e^{-t} + 2e^{-t}(-\sin t + \cos t)$$

$$= -2e^{-t} \times 2 \sin t \quad \boxed{\checkmark}$$

$$= -4e^{-t} \sin t$$

$$\ddot{x} = \sin t \times 4e^{-t} + -4e^{-t} \cos t$$

$$= 4e^{-t}(\sin t - \cos t) \quad \boxed{\checkmark}$$

$$(ii) \quad \text{As } t \rightarrow \infty, x \rightarrow 0 \text{ since } e^{-t} \rightarrow 0. \quad \boxed{\checkmark}$$

$$(iii) \quad 0 = 2e^{-t}(\cos t + \sin t) \text{ but } e^{-t} \neq 0$$

$$\cos t + \sin t = 0$$

$$\sin t = -\cos t \quad \boxed{\checkmark}$$

$$\tan t = -1$$

$$t = \frac{3\pi}{4}, \frac{7\pi}{4} \quad \boxed{\checkmark}$$

$$(iv) \quad \dot{x} = -4e^{-t} \sin t \text{ moving in a positive direction } \dot{x} > 0, \quad \boxed{\checkmark}$$

$$-4e^{-t} \sin t > 0$$

$$\sin t < 0$$

$$\pi < t < 2\pi$$

$$(v) \quad \text{Stationary when } \dot{x} = 0 \rightarrow t = 0, \pi, 2\pi \quad \boxed{\checkmark} \boxed{\checkmark}$$

$$(vi) \quad x = 2e^{-t}(\cos t + \sin t) \quad t = 0, \pi, 2\pi$$

$$x = 2(\cos 0 + \sin 0) = 2$$

$$x = 2e^{-\pi}(\cos \pi + \sin \pi) = -0.086$$

$$x = 2e^{-2\pi}(\cos 2\pi + \sin 2\pi) = 0.004 \quad \boxed{\checkmark} \boxed{\checkmark}$$

(vii)

